

EE236: Electronic Devices Lab

Lab 7: Bipolar Junction Transistor and Hetero-junction Bipolar Transistor

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1 Aim of the experiment

- Plotting output characteristics of BJT in CB configuration
- Frequency response of BJT vs HBT in CE configuration

2 Theory

BJT is a 3-terminal 2-junction transistor that is used in high-frequency applications (such as RF circuits). It has three terminals: Base (B), Collector (C), and Emitter (E). A BJT allows a small current injected at its Base to control a much larger current flowing between the Emitter and Collector terminals, making the device capable of amplification and switching.

Regions of Operation of BJT (NPN):

B-E Junction	B-C Junction	Region of Operation
Reverse	Reverse	Cut-off
Forward	Reverse	Active
Reverse	Forward	Inverse-active
Forward	Forward	Saturation

Table 1: Regions of Operation of NPN BJT

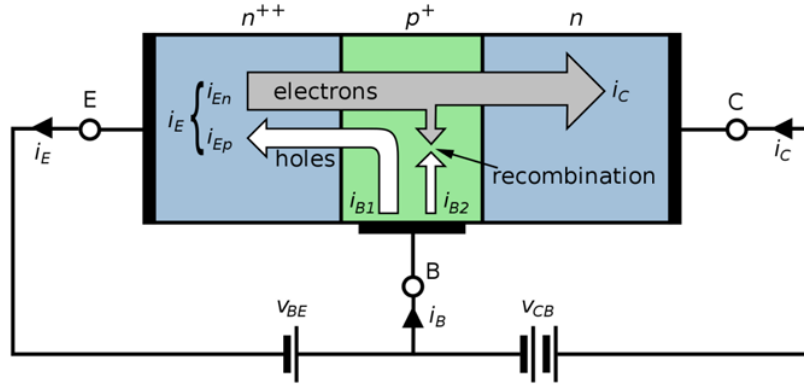


Figure: Current mechanisms in NPN BJT, Active Region

Base Transport Factor (α_T): The fraction of the minority carriers injected into the base that successfully diffuse across the width of the base and enter the collector.

$$\alpha_T = \frac{i_C}{i_E}$$

Emitter Efficiency (γ):

$$\gamma = \frac{i_{E_n}}{i_E}$$

Common Emitter Current Gain (β):

$$\beta = \frac{i_C}{i_B}$$

3 Components required

The following components are needed in order to perform the experiment:

- BC547 BJT
- MT3S1 HBT
- Resistors (PART 1): 1k Ω , 470 Ω

- Resistors (PART 2): $15\text{k}\Omega$, $18\text{k}\Omega$, $10\text{k}\Omega$, $1.2\text{k}\Omega$, 250Ω
- Potentiometer: $1\text{k}\Omega$
- Capacitor: $4.7\mu\text{F}$
- Breadboard, Multimeters, and connecting wires

4 Part-1: BJT Parameters in CB configuration

4.1 Circuit

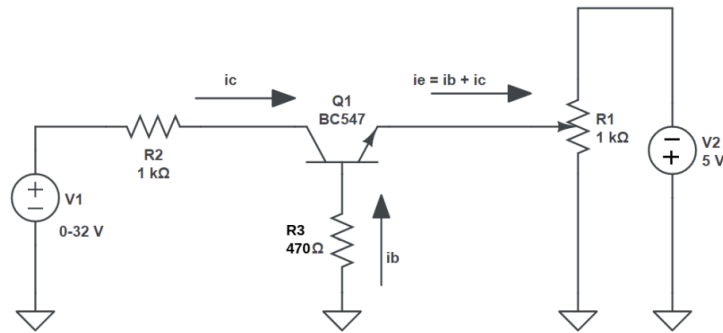


Figure 1: CB circuit

4.2 Readings

Table 1: $i_E = 3mA$

i_C (mA)	V_{cb} (V)
3.03	1.03
3.04	5.07
3.05	7.36
3.06	8.96
3.07	11.32
3.08	11.60
3.09	12.77

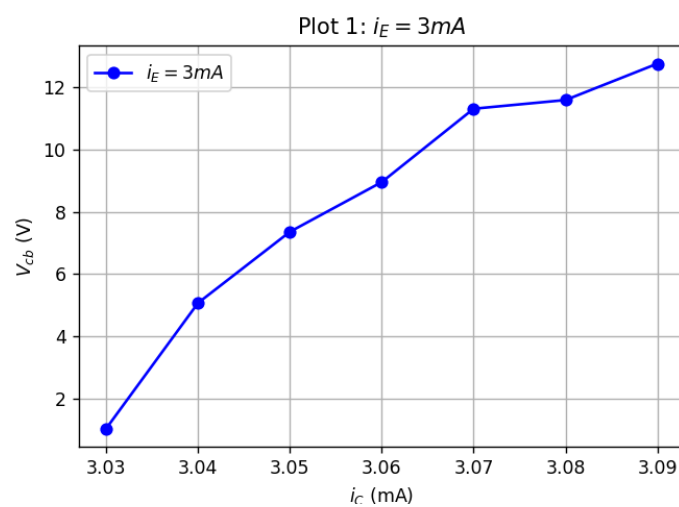


Table 2: $i_E = 6mA$

i_C (mA)	V_{cb} (V)
5.19	0.68
5.51	0.67
6.00	0.61
6.11	0.53
6.16	0.49
6.17	0.45
6.18	0.28
6.19	0.02

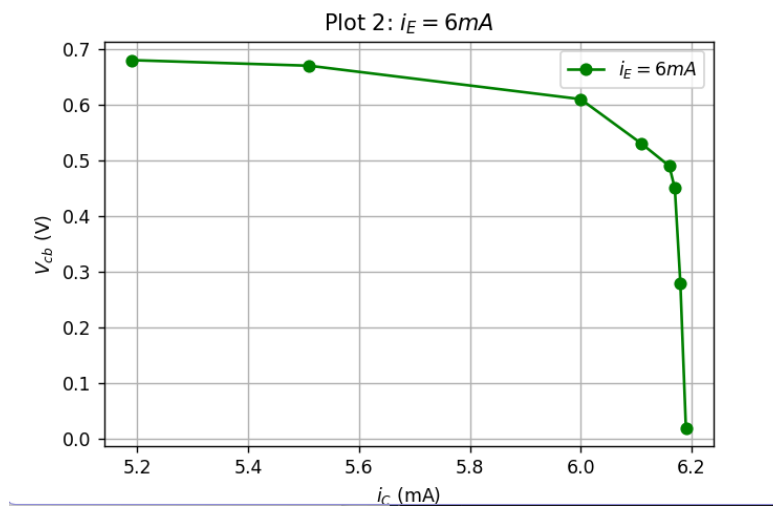


Table 3: $i_E = 9mA$

i_C (mA)	V_{cb} (V)
6.23	0.72
6.92	0.71
7.21	0.69
7.84	0.68
8.25	0.67
8.61	0.64
9.09	0.49
9.11	0.26
9.12	0.16

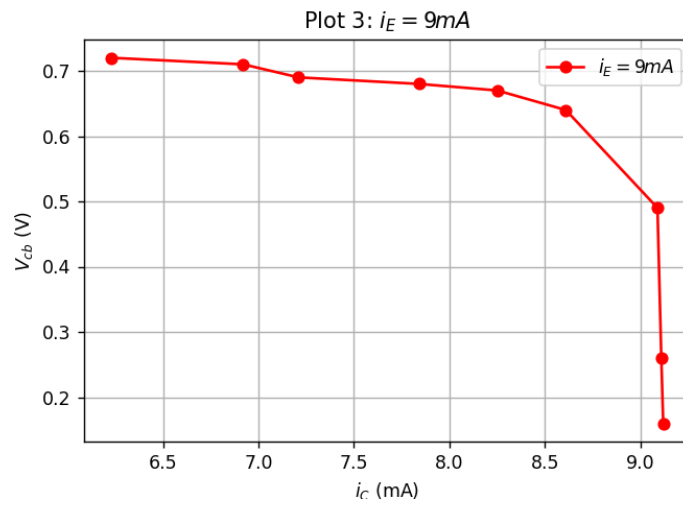
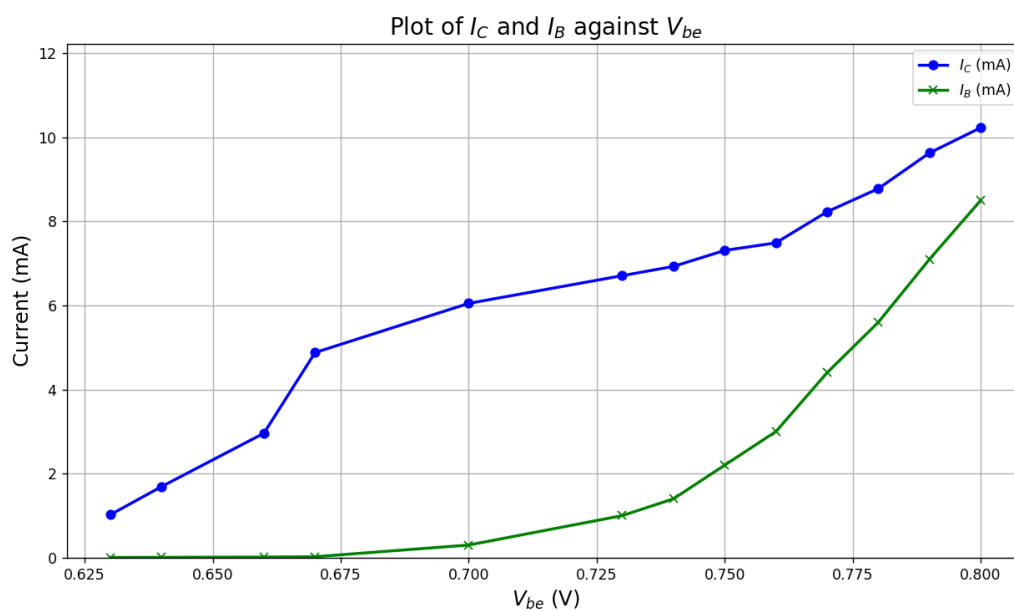


Table 4: Collector and base currents (I_C and I_B) against varying base emitter (V_{BE}) voltage at a fixed collector to base bias voltage (V_{CB})

V_{be} (V)	I_C (mA)	I_B (mA)
0.63	1.02	0.004
0.64	1.69	0.007
0.66	2.96	0.013
0.67	4.88	0.022
0.70	6.05	0.3
0.73	6.71	1.0
0.74	6.93	1.4
0.75	7.31	2.2
0.76	7.49	3.0
0.77	8.23	4.4
0.78	8.78	5.6
0.79	9.63	7.1
0.80	10.23	8.5



5 Part-2: Frequency response of BJT vs HBT

5.1 BJT: BC547

5.1.1 Circuit

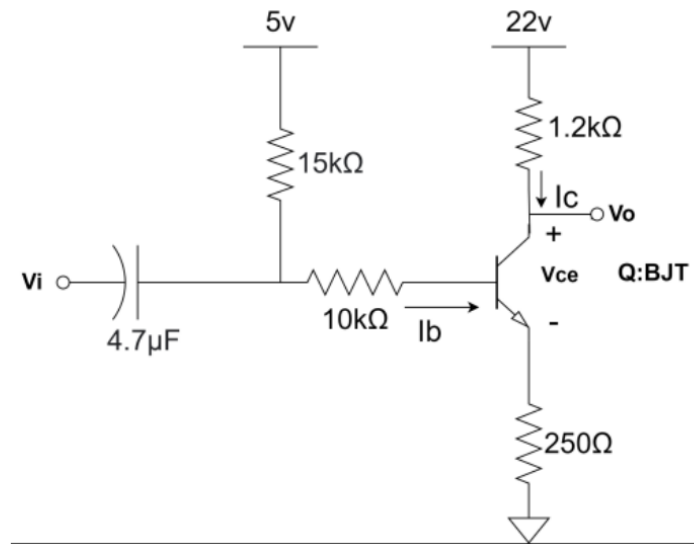


Figure 2: BJT CE Circuit

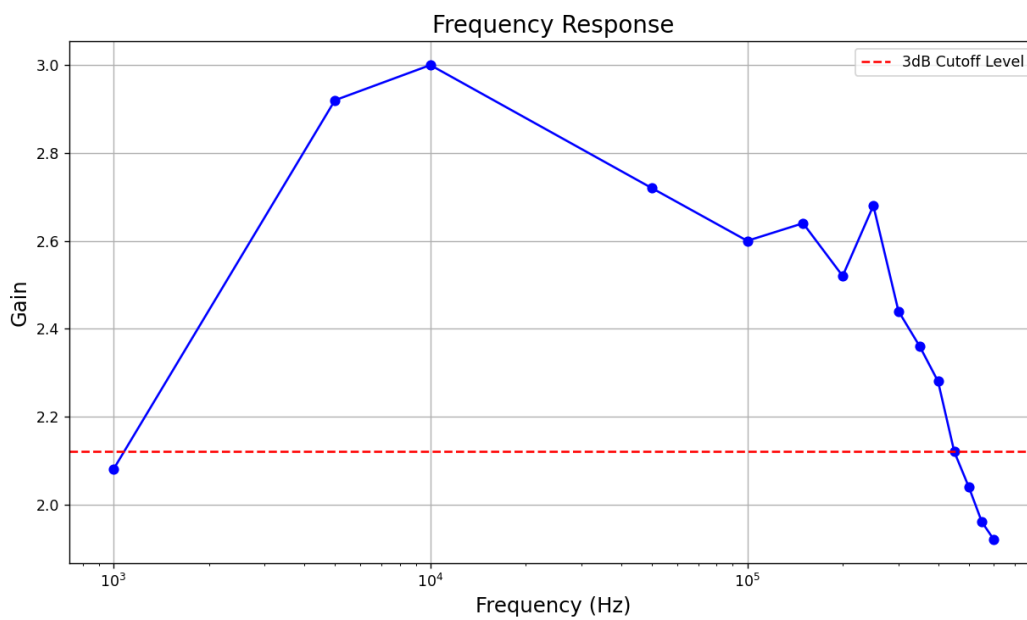
The DC bias operating point in the Common Emitter circuit was fixed by:

$$V_{CE} = 6.0\text{ V}, \quad I_C = 12\text{ mA}, \quad I_B = 50\mu\text{A}.$$

Input signal: 500 mV , peak to peak

5.1.2 Readings

Frequency (Hz)	V_{out} (pk-pk) (V)	Gain
1k	1.1	2.08
5k	1.46	2.92
10k	1.5	3.00
50k	1.36	2.72
100k	1.34	2.60
150k	1.32	2.64
200k	1.26	2.52
250k	1.24	2.68
300k	1.22	2.44
350k	1.18	2.36
400k	1.14	2.28
450k	1.06	2.12
500k	1.02	2.04
550k	0.98	1.96
600k	0.96	1.92
3dB cutoff frequency = 450000 Hz		



5.2 HBT: MT3S11

5.2.1 Circuit

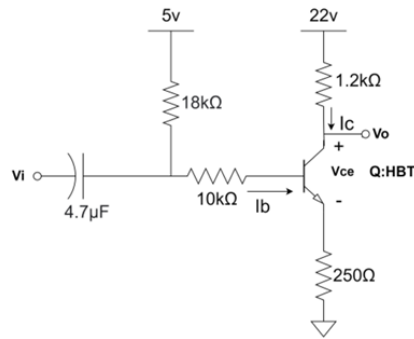


Figure: HBT CE Circuit

Figure 3: HBT CE Circuit

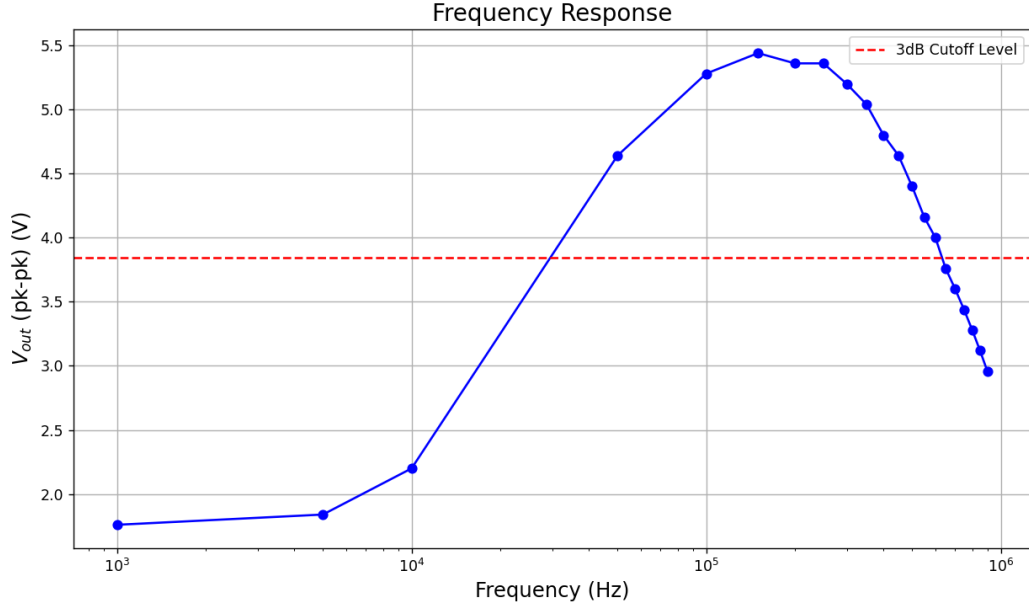
The DC bias operating point in the Common Emitter circuit was fixed by:

$$V_{CE} = 3.5V, \quad I_C = 12\text{ mA}, \quad I_B = 50\text{ }\mu\text{A}.$$

Input signal: 500 mV, peak to peak

5.2.2 Readings

Frequency (Hz)	V_{in} (mV)	V_{out} (pk-pk) (V)
1k	512	1.76
5k	512	1.84
10k	512	2.20
50k	512	4.64
100k	504	5.28
150k	504	5.44
200k	504	5.36
250k	504	5.36
300k	512	5.20
350k	512	5.04
400k	512	4.80
450k	512	4.64
500k	512	4.40
550k	520	4.16
600k	520	4.00
650k	520	3.76
700k	520	3.60
750k	520	3.44
800k	520	3.28
850k	520	3.12
900k	520	2.96
3dB cutoff frequency = 625000 Hz		



5.3 Comparison of 3-dB Frequency of HBT & BJT

5.3.1 3-dB Frequency Values

- **HBT:** 625,000 Hz (625 kHz)
- **BJT:** 450,000 Hz (450 kHz)

The 3-dB frequency, also known as the cutoff frequency, represents the point at which the output power drops to half of its maximum value, equivalent to a drop of 3 dB in voltage gain. In this comparison, the HBT demonstrates a superior 3-dB frequency compared to the BJT.

5.3.2 Superior Performance of HBTs at Higher Frequencies

Heterojunction Bipolar Transistors (HBTs) demonstrate enhanced performance at high frequencies due to several critical factors:

1. **Material Composition:** HBTs utilize different semiconductor materials for the emitter and base regions, creating a heterojunction. This design restricts hole movement from the base to the emitter, allowing for a higher doping density in the base without compromising gain. This results in improved charge carrier transport and reduced recombination losses [1].

2. **Reduced Capacitance:** The structure of HBTs minimizes the collector-to-substrate capacitance, which is crucial for high-frequency operation. Lower capacitance allows for faster switching speeds and improved frequency response [3].

3. **Higher Emitter Injection Efficiency:** The heterojunction design enhances emitter injection efficiency by blocking hole injection from the base into the emitter. This characteristic enables HBTs to maintain high current gain even at elevated frequencies [2].

4. **Lower Base Resistance:** The ability to achieve a heavily doped base in HBTs leads to lower base resistance compared to BJTs. This reduction contributes to faster response times and better high-frequency performance [1].

These factors collectively enable HBTs to operate efficiently at frequencies that exceed those achievable by traditional BJTs, making them ideal for applications in high-speed communications and RF systems.

References

- [1] Y.C. Chou and R. Ferro, "Heterojunction Bipolar Transistors," JPL, NASA, 2024.
- [2] "Heterojunction Bipolar Transistors," Cadence, March 22, 2024.
- [3] "Heterojunction bipolar transistor," Wikipedia, accessed October 2024.